

Your Power BI Report is Only as Good as Its Data Model



Why Data Modeling Matters

- Foundation for all Power BI reports and dashboards
- Directly impacts performance and speed
- Ensures accurate calculations and metrics
- Simplifies the user experience for report consumers
- Makes maintenance and changes easier later
- Turns raw data into a trusted analytics asset



Meet the Star Schema

- The gold standard for Power BI data models
- Central fact table surrounded by dimension tables
- Fact tables hold measurable data (e.g., sales amount, quantity)
- Dimension tables hold descriptive data (e.g., product, date, customer)
- Designed for high performance and simplicity
- Resembles a star in the relationship view



The Snowflake Schema

- A variation of the star schema
- Dimension tables are normalized into multiple related tables
- Can resemble a snowflake in the relationship view
- May be necessary for complex, hierarchical dimensions
- Generally not recommended for Power BI
- Can lead to more complex models and DAX



Star vs Snowflake: Choose Wisely

- Prefer Star Schema for most Power BI scenarios
- Star is simpler for end-users to understand
- Star typically delivers better performance
- Snowflake might be used for very large dimensions
- Avoid snowflaking unless you have a specific reason
- Always denormalize dimensions into single tables when possible



Relationship Best Practices

- Use single-direction, filter cross-filtering
- Avoid bidirectional relationships where possible
- Always define the correct cardinality (one-to-many, etc.)
- Hide unnecessary fields from report view
- Ensure active relationships are set correctly
- Use descriptive table and column names



Common Modeling Mistakes

- Importing every available column from a source
- Creating a single large, flat table
- Using unnecessary bidirectional relationships
- Leaving auto-date/time tables enabled
- Not setting a date table for time intelligence
- Building a model without business user input



Pro Tips for Success

- Start with a clear business question
- Collaborate with business stakeholders on the design
- Use a consistent and clear naming convention
- Implement role-playing dimensions correctly
- Leverage the Model View to organize tables
- Document your model for future users



Your Action Plan

- Audit your existing models for star schema compliance
- Review and simplify complex relationships
- Eliminate any unnecessary columns from your tables
- Set a designated date table for your time intelligence
- Practice building a new model from scratch
- Share your knowledge with your team



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